44. ACID-BASE BALANCE AND THE HYPOTHALAMIC-PITUITARY SYSTEM

DURATION : 04:50

STUDY SHEET

COURSE SUMMARY

This video delves into the interaction between acid-base balance and the hypothalamic-pituitary system, highlighting the crucial role of the hypothalamus in regulating body tone. Through concepts such as ergotropy and trophotropy, you will learn how acid-base variations influence not only your posture but also your general well-being. Synaptic transmission and its integration at higher neurological levels are explored, offering an understanding of bodily responses and sphincters. By observing these interactions, you will be able to better adapt your practices in osteopathy or manual therapy, allowing for an integrative approach to the body and its chemistry.

ACID-BASE BALANCE AND THE HYPOTHALAMIC-PITUITARY SYSTEM

We will delve deeper into understanding the body in relation to its **acid-base balance**, its **nervous system**, and the chemistry of its tone.

At a certain point in its development, the body integrates a **thalamic** and **hypothalamic** level into its centre.

ROLE OF THE HYPOTHALAMUS IN REGULATION

The **hypothalamus** plays a crucial role, at a deep neurological level, in performing **regulation**. This regulation redistributes a preferential tone, of either an **ergotrophic** or **trophotrophic** type.

Cellular information, which arrives via an **oxidation-reduction** system or **cyclic AMP** and **cyclic GMP**, is integrated at the level of the hypothalamus. The latter then provides feedback.

To simplify, your acid-base terrain will be integrated, via the bloodstream, at the **hypothalamic-pituitary** level. This will transmit information related to your **limbic system** and your **Papez circuit**, re-integrating this into a tonality.

Thus, it will give you a side:

- **Ergotrophic**: oriented towards work, **orthosympathetic**.

- **Trophotrophic**: oriented towards recovery.

This manifests as a posture. Through sympathetic or parasympathetic tone, the hypothalamus must be informed of the acid-base balance, expressing a **trophotrophic** or **ergotrophic** tone.

INTEGRATION OF ACID-BASE BALANCE

At the level of the hypothalamus, a predominant **acidosis** or **alkalosis** between tissues manifests. These are local adaptations integrated into intermediate, **diaphragmatic** levels, connected to continuously stimulated movements.

This means that we have **synaptic transmission** that carries **cyclic AMP** or **gonadocyclic** type information. These two types of transmission are expressed at the cellular level by a dynamic system, i.e., an oxidation-reduction system, or in other words, acid-base.

The whole is recovered at the neurological level, integrated into the **corticospinal pathway** and the **limbic pathway**. All memories, including your diet, are integrated at this level and provide a response.

BODY RESPONSE AND POSTURE

This response manifests in your system via the **paraventricular** or **supraoptic nuclei**, leading to a bodily response.

For example, you can go into a **left-right** or **right-left spiral**, and this will influence your **sphincters**.

At the level of your **pancreatic facial pentagon**, you have different areas with specific sphincters:

- The **pyloric sphincter**.
- The **duodenal sphincter**.
- The **ileocecal sphincter**.
- The **recto-anal sphincter**.

These sphincters integrate according to your movement. If you shift your centre of gravity to the right or to the left, you will have a different action on these sphincters.

If you are blocked, this affects your sphincters, which can be either open or closed. You can be in **diarrhoea** or **constipation**, or in a **precancerous** or **pro-inflammatory** state. All of this, whether you are in **para** or **ortho**, has repercussions in specific chemistry.

The body tries to balance the whole and to equilibrate itself, putting you in a state of constipation or encouraging you to open your centre and have diarrhoea.

All of this is expressed by this chemistry integrated at the hypothalamic level.

We can observe in your **chemo-postural** plan phases of **external rotation** and phases of **internal rotation**.

Observation of posture and balance allows us to read the body's chemistry. This is why the body continuously tries, through the respiratory act, to adapt to the environment programmed by its hypothalamic-pituitary system, located at the base of the skull.